Measure-Preserving Equivalence Relations

Definition:

- Given a Borel relation $R \subseteq X \times Y$, we define the sections

$$R_x = \{y \in Y : (x, y) \in R\}$$
$$R^y = \{x \in X : (x, y) \in R\}.$$

- We say a relation $Q \subseteq X \times Y$ is a *graph* if, whenever (x, y) and (x, y') are in Q, one has $y = y'$. Equivalently, a relation $Q \subseteq X \times Y$ is a graph if and only if it is equal to $\text{graph}(f)$ for some partial function $f : \text{dom}(f) \to \text{range}(f)$.
- Given a relation $R \subseteq X \times Y$, the projection $\text{proj}_X(R)$ is defined by

$$\text{proj}_X(R) = \{x \in X : \text{ there exists } y \in Y \text{ such that } (x, y) \in R\}.$$

We will black box the following two theorems. The proofs require deep results in descriptive set theory.

Theorem (Lusin–Novikov Theorem)**:** Let $K \subseteq X \times Y$ be a relation. If every section K_x is countable, then $\text{proj}_X(K) \subseteq X$ is Borel, and there is a $Q \subseteq K$ such that $\text{proj}_X(Q) = \text{proj}_X(K)$. Equivalently, there is a Borel partial function $f : \text{proj}_X(K) \to Y$ such that $\text{graph}(f) \subseteq K$ (such a function is called a *uniformization* for the relation).

Moreover, K is a countable union of all such Borel graphs.

Theorem (Lusin–Suslin Theorem)**:** Let $f : X \to Y$ be a map of standard Borel spaces. If $A \subseteq X$ is Borel and $f|_A$ is injective, then $f(A)$ is Borel, and f is a Borel isomorphism of A onto $f(A)$.

Remark: Since any pair of standard Borel spaces can be equipped with a zero-dimensional Polish topology making a given Borel map continuous, the proof proceeds via the reduction to the continuous case.

Furthermore, any uncountable standard Borel is Borel isomorphic to $2^{\mathbb{N}}$.

Lemma (Marker Lemma)**:** Every aperiodic (i.e., has infinite classes) countable Borel equivalence relation E admits a sequence of subsets $(B_n)_n$ of complete sections with $B_n \supseteq B_{n+1}$ and $\bigcap_{n\in\mathbb{N}} B_n = \emptyset$. Such a sequence is called a *vanishing sequence of markers*.

Proof. Assume that the equivalence relation E is on $X = 2^{\mathbb{N}}$. Let $s_n(x)$ be the lexicographically smallest $s \in 2^n$ such that $\left|[x]_E \cap \mathcal{N}_s\right| = \infty$, where $\mathcal{N}_s$ is the Borel cylinder set for s.

Say $x \in A_n$ if and only if $x|_n = s_n(x)$. Then, $\{A_n : n \in \mathbb{N}\}$ is a decreasing sequence of Borel sets which intersects each class of E in infinitely many points. It follows that

$$A = \bigcap_{n\in\mathbb{N}} A_n$$

intersects each class of E in at most one point. Therefore, $B_n = A_n \setminus A$ is our desired sequence of markers. □

Definition: Let (X, μ) be a standard probability space.

- Given a Borel equivalence relation $R \subseteq X \times X$, the *full group* of R is the set of all Borel automorphisms $f : X \to X$ such that $\text{graph}(f) \subseteq R$.
- The set $[[R]]$ consists of all partial automorphisms $\phi : \text{dom}(\phi) \to \text{range}(\phi)$.
- The equivalence relation R is said to measure-preserving if for all $\phi \in [[R]]$, we have $\mu(\text{dom}(\phi)) = \mu(\text{range}(\phi))$.
- If $A \subseteq X$ is any subset, then the saturation $[A]_R$ is given by

$$[A]_R = \bigcup_{x\in A} [x]_R.$$

- We say A is R-invariant if $\mu(A \triangle [A]_R) = 0$. A measure-preserving relation R is said to be ergodic if any R-invariant subset is either null or conull.

Remark: Given a Borel map $f : R \to \mathbb{C}$, we define the action of $T \in [[R]]$ by $Tf(x,y) = f\left(T^{-1}.x, y\right)$ if $x \in \text{range}(T)$ and $Tf(x,y) = 0$ otherwise.

Analogously, if $f : X \to \mathbb{C}$ is a Borel map, we let $Tf(x) = f\left(T^{-1}.x\right)$ if $x \in \text{range}(T)$ and $Tf(x) = 0$ otherwise.

Finally, if $A \subseteq X$ is Borel, then $\text{id}_A \in [[R]]$ is the partial transformation with domain A and is the identity map on A.

The most important case of a measure-preserving equivalence relation is the *orbit equivalence relation* of a probability measure-preserving action $G \curvearrowright (X, \mu)$. The relation is given by saying $x \sim y$ if there exists some $s \in G$ such that $y = s.x$.

Two probability measure-preserving actions of groups $G \curvearrowright (X, \mu)$ and $G \curvearrowright (Y, \nu)$ are said to be orbit equivalent if there is a G-invariant conull set $X' \subseteq X$, an H-invariant conull set $Y' \subseteq Y$, and a Borel isomorphism $\varphi : X' \to Y'$ satisfying $\varphi(G.x) = H.\varphi(x)$. One observes that, given a probability measure-preserving action $G \curvearrowright (X, \mu)$, the G-invariance of any subset only depends on the orbit equivalence relation, so a probability measure-preserving action is ergodic if and only if its orbit equivalence relation is ergodic.

Furthermore, one can show that a countable Borel relation is the orbit equivalence relation of some group (not necessarily that of a freely acting group, though). This is the *Feldman–Moore theorem*.

Definition: Given a measure-preserving equivalence relation R on (X, μ), we say $Q \subseteq R$ is a *partial subequivalence relation* if Q is an equivalence relation defined on

$$\text{dom}(Q) = \{x \in X : (x,x) \in Q\}.$$

A genuine subequivalence relation $Q \subseteq R$ is a partial subequivalence relation with $\text{dom}(Q) = X$.

Given a (countable) measure-preserving equivalence relation on (X, μ). Define a measure on the Borel subsets $A \subseteq R$ by

$$\omega(A) = \int_X |A_x| \, d\mu(x).$$

Then, we observe that ω is a σ-finite measure as follows.

- Using the Lusin–Novikov Theorem, the relation R is a countable union of Borel graphs, which we will write as $R = \bigcup_n G_n$. Notice that if $Q \subseteq R$ is any such Borel graph, then $|Q_x| = 0$ or $|Q_x| = 1$ depending on if $x \in \text{proj}_X(Q)$.
- For any relation E, let

  $$E^{-1} = \{(y,x) : (x,y) \in E\}.$$

  Each set of the form

  $$G_n' = G_n \setminus \bigcup_{m<n} G_m$$

  is a graph of a partial function contained in R (one that is countable-to-one). Since R is an equivalence relation, the sections $\left((G_n')^{-1}\right)_x = (G_n')^x$ are also countable, so we may split G_n' into countably many disjoint pieces $H_{k,n}$ corresponding to partially defined maps $X \to X$ that are one-to-one (i.e., partially defined transformations $X \to X$).
- Reindexing the $H_{k,n}$ to $\{B_n : n \in \mathbb{N}\}$, we define $f_n = \pi_1|_{B_n}$ to be the restriction of the first coordinate projection to B_n. Each f_n is continuous and injective, so it is a Borel isomorphism onto its image. If we define

  $$\omega_n(A) = \mu(f_n(A \cap B_n)),$$

then each ω_n is a finite Borel measure on R, following from the fact that μ is a probability measure.

- Finally, the function $x \mapsto |A_x|$ is given by $\sum_n \chi_{f_n(A \cap B_n)}$, and is Borel.

Therefore, ω is a well-defined σ-finite measure.

Amenability and Hyperfiniteness

Definition: Let R be a countable Borel equivalence relation.

(i) The relation R is *amenable* if there is a unital positive linear map $\varphi\colon L^\infty(R,\omega) \to L^\infty(X)$ satisfying $\varphi(T.f) = T.\varphi(f)$ for all $T \in [[R]]$ and $f \in L^\infty(R,\omega)$.

(ii) The relation R is said to satisfy *Reiter's property* if there is a sequence of positive Borel functions $\lambda^n\colon R \to \mathbb{C}$ such that for each $y \in X$, the map φ_y^n on $[y]_R$ given by $\varphi_y^n(x) = \lambda^n(x,y)$ is a norm-one element of $\ell^1\big([y]_R\big)$ and $\left\|\lambda_y^n - \lambda_{y'}^n\right\|_{\ell^1} \xrightarrow{n\to\infty} 0$ for ω-a.e. $(y,y') \in R$.

(iii) We say the relation R is *hyperfinite* if it is a.e. equal to the union of an increasing sequence of finite Borel subequivalence relations.

Notice that if we write the C^*-algebra $L^\infty(X)$ as $C(\Sigma)$ for some pathological compact space Σ, then composing point evaluation with the unital positive linear functional $\varphi\colon L^\infty(R,\omega) \to L^\infty(X)$ is a contractive homomorphism, so φ is itself contractive.

Theorem (Slaman–Steel and Weiss)**:** Let R be a countable Borel equivalence relation. Then, the following are equivalent:

(i) R is the countable union of an increasing sequence $R_1 \subseteq R_2 \subseteq \cdots$ of finite Borel subequivalence relations;

(ii) R is the orbit equivalence relation of a Borel $\mathbb{Z}$-action.

Suppose R is a probability measure-preserving equivalence relation on (X,μ). If R is amenable, and $\varphi\colon L^\infty(R,\omega) \to L^\infty(X)$ witnesses the amenability, then we may define a unital positive linear functional $\sigma\colon L^\infty(R,\omega) \to \mathbb{C}$ by

$$\sigma(f) = \int_X \varphi(f)\, d\mu.$$

Notice then that $\sigma(T.f) = \sigma\Big(\mathrm{id}_{\mathrm{dom}(T)}.f\Big)$ for all $T \in [[R]]$ and $f \in L^\infty(R,\omega)$. Such a linear functional is called an invariant state on $L^\infty(R,\omega)$. The existence of an invariant state is a weakened condition of amenability.

Given Borel subsets $A, B \subseteq X$, write

$$R_A = R \cap (A \times A)$$
$$R_{A,B} = R \cap (A \times B).$$

If A is not null, we may view R_A as a probability measure-preserving equivalence relation upon normalizing the measure of A, and similarly, we may consider invariant states on $L^\infty\Big(R_A, \mu(A)^{-1}\omega\Big)$. We say two Borel subsets $A, B \subseteq X$ are equivalent if there is $T \in [[R]]$ such that $\mathrm{dom}(T) = A$ and $\mathrm{range}(T) = B$.

Lemma: Let A and B be Borel subsets of X. If $\mu([B]_R \cap A) > 0$, then there is a nonnull Borel subset of B that is equivalent to a subset of A.

Proof. By multiple applications of the Lusin–Novikov theorem (akin to how we proved that ω was a σ-finite measure), we may find a countable partition R_n of $R_{A,B}$ such that each R_n is the graph of some $T_n \in [[R]]$. Then,

$$[B]_R \cap A = \bigcup_{n=1}^{\infty} \mathrm{dom}(T_n).$$

Since $\mu([B]_R \cap A) > 0$, $\mu(\text{dom}(T_n)) > 0$ for some $n \in \mathbb{N}$, and $\text{range}(T_n) \subseteq B$ has measure $\mu(\text{dom}(T_n))$ equivalent to $\text{dom}(T_n) \subseteq A$. □

Lemma: Let A and B be Borel subsets of X. There exists a Borel subset $B_1 \subseteq B$ such that B_1 is equivalent to a subset of A and $\mu([B \setminus B_1]_R \cap A) \leq \mu(B)$.

Proof. Let $\mathcal{R}$ be the collection of all families of Borel subsets $\left\{R_j : j \in J\right\}$ such that each R_j is non-null and $R_j = \text{graph}\left(T_j\right)$ for some family of maps $\left\{T_j : j \in J\right\}$ with each of $\left\{\text{range}\left(T_j\right) : j \in J\right\}$ and $\left\{\text{dom}\left(T_j\right) : j \in J\right\}$ pairwise disjoint.

Say $\left\{R_j : j \in J\right\} \leq \left\{R'_{j'} : j' \in J'\right\}$ if $J \subseteq J'$ and $R_j = R'_j$ for all $j \in J$. By Zorn's Lemma, there is a maximal element which we will denote $\left\{R_j : j \in J\right\}$. Let

$$A_1 = \bigcup_{j \in J} \text{dom}\left(T_j\right)$$
$$B_1 = \bigcup_{j \in J} \text{im}\left(T_j\right).$$

Since $\bigcup_{j \in J} R_j$ is the graph of some element in $[[R]]$, we have that A_1 and B_1 are equivalent. If it were the case that $\mu([B \setminus B_1]_R, A \setminus A_1) > 0$, then there would be some nonnull Borel subset of $B \setminus B_1$ equivalent to a subset of $A \setminus A_1$ via some map $T \in [[R]]$; adding the graph of T to $\left\{R_j : j \in J\right\}$ would contradict maximality. Therefore, we have

$$\mu([B \setminus B_1] \cap (A \setminus A_1)) = 0$$

so

$$\begin{aligned} \mu([B \setminus B_1]_R, A) &\leq \mu(A_1) \\ &= \mu(B_1) \\ &\leq \mu(B). \end{aligned}$$

□

Lemma: Let A and B be Borel subsets of X with $B \subseteq [A]_R$, $\varepsilon > 0$.

Then, there is a finite collection $\{B_k : k = 1, \ldots, n\}$ of pairwise disjoint Borel subsets of B such that each B_j is equivalent to a subset of A and $\mu\left(B \setminus \bigcup_{k=1}^{n} B_k\right) < \varepsilon$.

Proof. Let $\mathcal{R}$ be the family of collections of pairwise disjoint Borel subsets $\left\{B_j : j \in J\right\}$ with each B_j nonnull and equivalent to a subset of A. Define a partial order on the family by saying that $\left\{B_j : j \in J\right\} \leq \left\{C_{j'} : j' \in J'\right\}$ if $J \subseteq J'$ and $B_j = C_j$ for all $j \in J$.

Zorn's Lemma gives a maximal such family, which we denote that $\left\{B_j : j \in J\right\}$. We observe that if $\mu\left(B \setminus \bigcup_j B_j\right) > 0$, then $B \setminus \bigcup_{B_j}$ has a nonnull Borel subset equivalent to a subset of A, which would contradict the maximality of $\left\{B_j : j \in J\right\}$. Thus $\mu\left(B \setminus \bigcup_j B_j\right) = 0$. Since each B_j is nonnull and the B_j are pairwise disjoint, J is countable. Let $J' \subseteq J$ be finite and $\mu\left(\bigcup_{j \in J \setminus J'} B_j\right) < \varepsilon$. Then, we have

$$\begin{aligned} \mu\left(B \setminus \bigcup_{j \in J'} B_j\right) &= \mu\left(\bigcup_{j \in J \setminus J'} B_j\right) \\ &< \varepsilon. \end{aligned}$$

□

Lemma: Let A and B be Borel subsets of X. Then, there is a sequence $\{B_n : n \in \mathbb{N}\}$ of pairwise disjoint Borel subsets of B such that each B_n is equivalent to a subset of A with

$$\lim_{n\to\infty} \mu\left(\left[B \setminus \bigcup_{j=1}^{n} B_j\right]_R \cap A\right) = 0.$$

Proof. Replacing B with $[A]_R \cap B$, we assume that $B \subseteq [A]_R$.

We will show that for any $\varepsilon > 0$, there is a finite collection $\{B_i : i \in I\}$ of pairwise disjoint Borel subsets of B such that each B_i is equivalent to a subset of A, and

$$\mu\left(\left[B \setminus \bigcup_{i\in I} B_i\right]_R \cap A\right) < \varepsilon.$$

By the previous lemma, we have a finite collection of subsets $\left\{B_j : j \in J\right\}$ of pairwise disjoint Borel subsets of B such that each B_j is equivalent to a subset of A and $\mu\left(B \setminus \bigcup_{j\in J} B_j\right) < \varepsilon$.

We may then find a Borel subset C of $B \setminus \bigcup_{j\in J} B_j$ such that C is equivalent to a subset of A with

$$\begin{aligned}\mu\left(\left[B \setminus \left(C \cup \bigcup_{j\in J} B_j\right)\right]_R \cap A\right) &\leq \mu\left(B \setminus \bigcup_{j\in J} B_j\right) \\ &< \varepsilon.\end{aligned}$$

Set $\{B_i : i \in I\} = \left\{B_j : j \in J\right\} \cup \{C\}$.

Recursively, for each $k = 1, 2, \ldots$, we may find a finite collection $\{B_i : i \in I_k\}$ of pairwise disjoint Borel subsets of $B \setminus \bigcup_{j=1}^{k-1} \bigcup_{i\in I_j} B_i$ such that each B_i is equivalent to a subset of A and

$$\mu\left(\left[B \setminus \bigcup_{j=1}^{k} \bigcup_{i\in I_k} B_i\right]_R \cap A\right) < 1/k.$$

Setting $n_k = \bigcup_{j=1}^{k} \left|I_j\right|$, and we list out $\{B_i : i \in I_k\}$ as $B_{n_{k-1}+1}, \ldots, B_{n_k}$. □

Lemma: Let $\sigma : L^\infty(R, \omega) \to \mathbb{C}$ be an invariant state such that $|\sigma(\mathrm{id}_B . f)| \leq \mu(B)\|f\|$ for all Borel $B \subseteq X$ and all $f \in L^\infty(R, \omega)$. Let A be a Borel subset of X with $\sigma\left(\chi_{R_A}\right) = 0$. Then, $\sigma\left(\chi_{R_{X,A}}\right) = 0$.